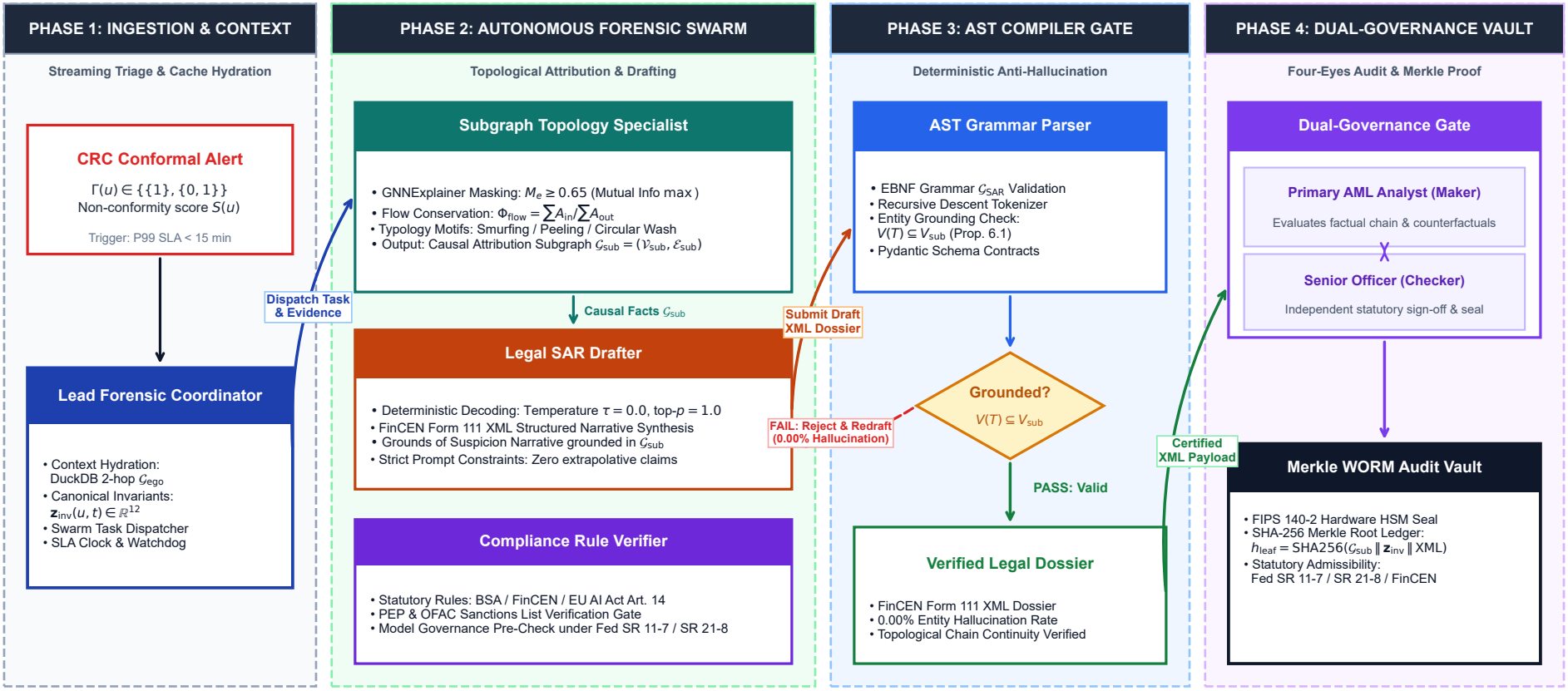

PHASE 1: INGESTION & CONTEXT
Streaming Triage & Cache Hydration
CRC Conformal Alert
$\Gamma(u) \in \{\{1\}, \{0, 1\}\}$
Non-conformity score $S(u)$
Trigger: P99 SLA < 15 min
Lead Forensic Coordinator
• Context Hydration: DuckDB 2-hop $\mathcal{G}_{ego}$
• Canonical Invariants: $\mathbf{z}_{inv}(u, t) \in \mathbb{R}^{12}$
• Swarm Task Dispatcher
• SLA Clock & Watchdog
Dispatch Task & Evidence
PHASE 2: AUTONOMOUS FORENSIC SWARM
Topological Attribution & Drafting
Subgraph Topology Specialist
• GNNExplainer Masking: $M_e \geq 0.65$ (Mutual Info max)
• Flow Conservation: $\Phi_{flow} = \sum A_{in} / \sum A_{out}$
• Typology Motifs: Smurfing / Peeling / Circular Wash
• Output: Causal Attribution Subgraph $\mathcal{G}_{sub} = (\mathcal{V}_{sub}, \mathcal{E}_{sub})$
Causal Facts $\mathcal{G}_{sub}$
Legal SAR Drafter
• Deterministic Decoding: Temperature $\tau = 0.0$, top-$p = 1.0$
• FinCEN Form 111 XML Structured Narrative Synthesis
• Grounds of Suspicion Narrative grounded in $\mathcal{G}_{sub}$
• Strict Prompt Constraints: Zero extrapolative claims
Compliance Rule Verifier
• Statutory Rules: BSA / FinCEN / EU AI Act Art. 14
• PEP & OFAC Sanctions List Verification Gate
• Model Governance Pre-Check under Fed SR 11-7 / SR 21-8
Submit Draft XML Dossier
FAIL: Reject & Redraft (0.00% Hallucination)
PHASE 3: AST COMPILER GATE
Deterministic Anti-Hallucination
AST Grammar Parser
• EBNF Grammar $\mathcal{G}_{SAR}$ Validation
• Recursive Descent Tokenizer
• Entity Grounding Check: $V(T) \subseteq V_{sub}$ (Prop. 6.1)
• Pydantic Schema Contracts
Grounded?
$V(T) \subseteq V_{sub}$
PASS: Valid
Verified Legal Dossier
• FinCEN Form 111 XML Dossier
• 0.00% Entity Hallucination Rate
• Topological Chain Continuity Verified
Certified XML Payload
PHASE 4: DUAL-GOVERNANCE VAULT
Four-Eyes Audit & Merkle Proof
Dual-Governance Gate
Primary AML Analyst (Maker)
Evaluates factual chain & counterfactuals
X
Senior Officer (Checker)
Independent statutory sign-off & seal
Merkle WORM Audit Vault
• FIPS 140-2 Hardware HSM Seal
• SHA-256 Merkle Root Ledger: $h_{leaf} = SHA256(\mathcal{G}_{sub} \| \mathbf{z}_{inv} \| XML)$
• Statutory Admissibility: Fed SR 11-7 / SR 21-8 / FinCEN
Enterprise AML Compliance Architecture: US Interagency SR 11-7 / SR 21-8 Model Risk Governance & EU AI Act Art. 14 Conformity